Aggregate Functions & Grouping

1. Count() Function; Write a SQL query to find the total number of employees in the company.

- SQL Query

```
SELECT COUNT(*) AS Total_employees
FROM employees;
```

Total_employees
10

2. Sum() Function; Write a SQL query to find the total salary paid to all employees in the IT department

- SQL Query

```
SELECT Sum('Salary') AS Total_Salary
FROM employees
WHERE ~~Salary~~ department = 'IT';
```

Total_Salary
220 000

3. V AVG() Function; Write a SQL query to calculate the average salary of employees in the HR department.

```
SELECT AVG(Salary) AS Average_Salary
FROM employees
WHERE department = 'HR';
```

Average_Salary
49,500

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4. MIN() and MAX() Functions; Write a SQL query to find the highest and the lowest ~~in the~~ salary in the company.

SQL Query

--- Calculating MIN()

```
SELECT MIN(Salary) AS (Lowest_Salary)
FROM employees;
```

Lowest_Salary
48000

SQL Query

--- Calculating MAX()

```
SELECT MAX(Salary) AS (highest_Salary)
FROM employees;
```

highest_Salary
62000

5. ~~Group BY and COUNT();~~ GROUP BY Statement; Write a SQL query to group employees by department and display the total salary paid in each department.

SQL Query

```
SELECT department,
       SUM(salary) AS total_salary
FROM employees
GROUP BY department;
```

department	total_Salary
IT	220 000
HR	99 000
Finance	119000
Marketing	105000

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6 GROUP BY and COUNT(); Write a SQL query to count how many employees work in each city.

SQL Query

```
SELECT COUNT(employees) AS total_employees
FROM employees
GROUP BY City;
```

City	total_employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. GROUP BY and ORDER BY; Write a SQL query to group employees by department, calculate the average salary in each department, and order the results in descending order of average salary.

SQL Query

```
SELECT department,
       AVG(salary) AS Average_salary
FROM employees
GROUP BY department
ORDER BY Average_salary DESC;
```

department	Average_Salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

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8. HAVING Clause; Write a SQL query to find departments where the total Salary paid exceeds 100,000. (use GROUP BY and HAVING)

SQL Query

```
SELECT department SUM(salary) AS total_Salary
FROM employees
GROUP BY department
HAVING COUNT(salary) > 100,000;
```

department	total_Salary
IT	220000
Finance	119000
Marketing	105000

9. Combining GROUP BY, HAVING, and ORDER BY

- Write a SQL query to list Cities where more than one employee works, ordered by the number of employees in descending order.

SQL Query

```
SELECT City COUNT(*) AS total_employees
FROM employees
GROUP BY City
HAVING COUNT(*) > 1
ORDER BY total_employees DESC;
```

City	total_employees
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

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10. Combining Aggregate Functions

- Write a SQL query to find the department with the highest average salary.

SQL Query

```
SELECT department, AVG(salary) AS average_salary
FROM employees
GROUP BY department
ORDER BY average_salary DESC
LIMIT 1;
```

department	average_salary
Finance	59500

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